

SUPPLEMENTAL MATERIAL

Data S1.

SUPPLEMENTAL METHODS

Additional information on LASSO and BKMR are provided below.

Least absolute shrinkage and selection operator (LASSO)

We applied the adaptive *least absolute shrinkage and selection operator* (LASSO) method to select important components from the PM_{2.5} elemental components that were associated with our outcome. The LASSO is a regression shrinkage and selection approach that can be used to select important predictors from a large list of correlated predictors.

The method applies a penalty to the regression coefficients by minimizing the sum of squared errors subject to the sum of the absolute values of the coefficients being less than a constant³⁰. The adaptive LASSO is an extension of the LASSO that uses weights for penalizing different coefficients in the penalty to achieve asymptotical normality and consistent selection.^{31,32}

In the adaptive LASSO model we included the fixed covariates that we adjusted for in the ordinary regression models outlined above. We applied the above penalty term to the PM_{2.5} components in the models, but not the adjustment variables. We selected the model with the minimum cross validation error, which yields a model result in which some coefficients are shrunk exactly to zero and some are non-zero. We then selected the components with non-zero coefficients. For this analysis we used the glmnet package in R version 3.5.3.

Bayesian kernel machine regression (BKMR).

Bayesian kernel machine regression (BKMR) is a joint estimation and variable selection approach that flexibly identifies important components within a multi-pollutant mixture. This method has been previously applied to assess the joint effect of in utero exposure to arsenic, manganese, and lead on children's neurodevelopmental outcomes²² and in a toxicology study of air pollution mixtures and hemodynamics³³

In this approach, the health outcome is regressed on a flexible function of the collection of components of interest. The form of the function is modeled via the use of a kernel, which is a matrix of pair-wise distances between each observation in the exposure space. BKMR³³ uses a Gaussian kernel function, which allows for the estimation of potentially nonlinear and non-additive exposure-response function for a set of correlated exposures. Taking a Bayesian approach to model fitting, the approach also incorporates a variable selection feature, which yields importance scores for each exposure defined as the posterior probability that an exposure should be included in the model. The resulting Bayesian credible intervals account for the uncertainty associated with both the selection of exposures in the model and estimation of the exposure-response function. In situations when mixture components are highly correlated, the data may not be able to distinguish among these correlated components, and one can apply a hierarchical variable selection approach, which incorporates information on the structure of the mixture into the model (grouped variable selection). From the fitted model, one can estimate a variety of summary statistics such as the posterior inclusion probability (PIP) for each of the components, as well as visualizations of different cross-sections of the exposure-response surface. This method is implemented in the BKMR R package (<https://jenfb.github.io/BKMR/overview.html>)³⁴.

The BKMR model is: $Y_i = h(X_i) + \beta_T Z_i + e_i$

where Z_i is the matrix of the fixed adjustment variables as defined in our main model, and X_i is the matrix of the components. The function h is an exposure–response function that allows for nonlinearity and/or interaction among the mixture components.

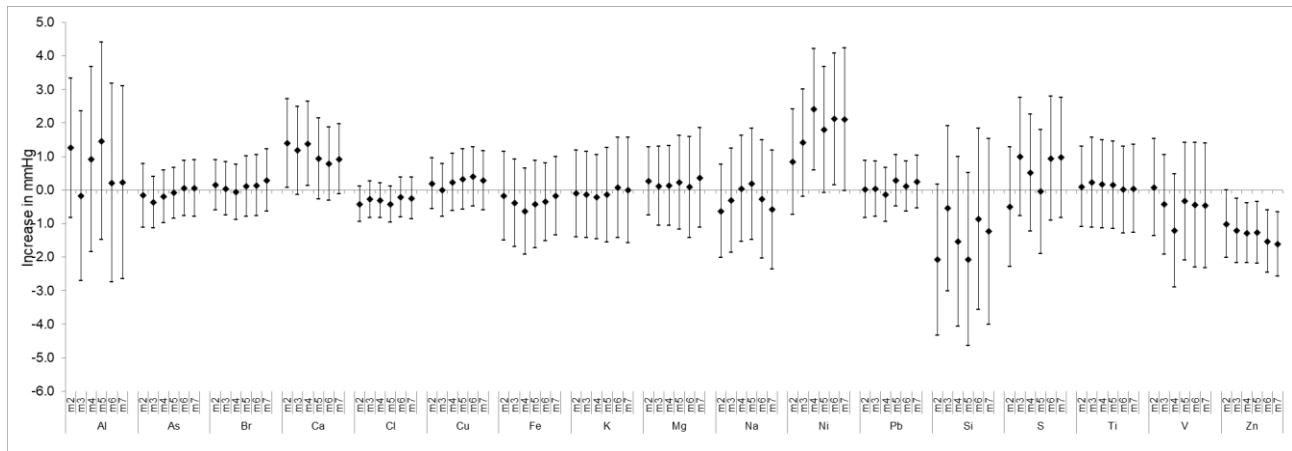
Table S1. Correlations between the PM2.5 components for the two days moving average.

	Two days moving averages																	
	PM _{2.5}	Al	Si	K	Ca	Ti	Fe	Mg	S	As	Cu	Zn	Br	Pb	V	Ni	Na	Cl
PM_{2.5}	1	0.34	0.16	0.60	0.22	0.31	0.33	0.46	0.78	0.17	0.17	0.14	0.37	0.21	0.27	0.16	0.48	-0.06
Al	0.34	1	0.93	0.58	0.57	0.73	0.61	0.51	0.43	0.09	0.13	0.02	0.14	0.22	0.05	-0.03	0.37	-0.05
Si	0.16	0.93	1	0.48	0.61	0.72	0.59	0.43	0.19	0.02	0.11	0.03	0.07	0.19	0.02	-0.05	0.22	-0.04
K	0.60	0.58	0.48	1	0.36	0.52	0.54	0.42	0.49	0.15	0.22	0.19	0.46	0.32	0.29	0.26	0.43	0.04
Ca	0.22	0.57	0.61	0.36	1	0.53	0.59	0.31	0.19	-0.01	0.20	0.16	0.14	0.21	0.24	0.12	0.26	0.10
Ti	0.31	0.73	0.72	0.52	0.53	1	0.66	0.40	0.31	0.06	0.22	0.15	0.23	0.19	0.12	0.06	0.23	-0.07
Fe	0.33	0.61	0.59	0.54	0.59	0.66	1	0.31	0.32	0.05	0.32	0.26	0.30	0.24	0.34	0.31	0.24	-0.05
Mg	0.46	0.51	0.43	0.42	0.31	0.40	0.31	1	0.57	0.15	0.08	0.06	0.29	0.24	0.18	0.07	0.71	0.16
S	0.78	0.43	0.19	0.49	0.19	0.31	0.32	0.57	1	0.24	0.13	0.10	0.35	0.19	0.22	0.11	0.65	-0.08
As	0.17	0.09	0.02	0.15	-0.01	0.06	0.05	0.15	0.24	1	0.04	-0.01	0.04	0.18	0.04	0.00	0.25	0.01
Cu	0.17	0.13	0.11	0.22	0.20	0.22	0.32	0.08	0.13	0.04	1	0.19	0.24	0.18	0.19	0.18	0.06	0.00
Zn	0.14	0.02	0.03	0.19	0.16	0.15	0.26	0.06	0.10	-0.01	0.19	1	0.26	0.29	0.26	0.29	0.06	0.05
Br	0.37	0.14	0.07	0.46	0.14	0.23	0.30	0.29	0.35	0.04	0.24	0.26	1	0.20	0.35	0.34	0.36	0.15
Pb	0.21	0.22	0.19	0.32	0.21	0.19	0.24	0.24	0.19	0.18	0.18	0.29	0.20	1	0.18	0.17	0.26	0.18
V	0.27	0.05	0.02	0.29	0.24	0.12	0.34	0.18	0.22	0.04	0.19	0.26	0.35	0.18	1	0.84	0.31	0.09
Ni	0.16	-0.03	-0.05	0.26	0.12	0.06	0.31	0.07	0.11	0.00	0.18	0.29	0.34	0.17	0.84	1	0.18	0.06
Na	0.48	0.37	0.22	0.43	0.26	0.23	0.24	0.71	0.65	0.25	0.06	0.06	0.36	0.26	0.31	0.18	1	0.37
Cl	-0.06	-0.05	-0.04	0.04	0.10	-0.07	-0.05	0.16	-0.08	0.01	0.00	0.05	0.15	0.18	0.09	0.06	0.37	1

Table S2. Correlations between the PM2.5 components for the seven days moving average.

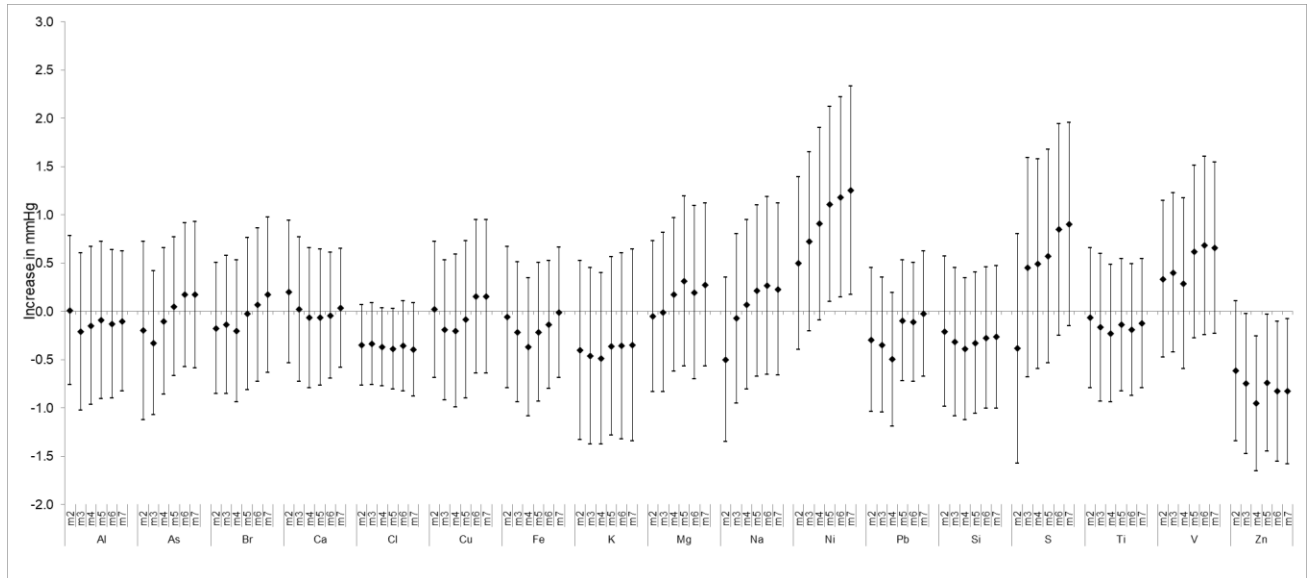
	Seven days moving averages																	
	PM _{2.5}	Al	Si	K	Ca	Ti	Fe	Mg	S	As	Cu	Zn	Br	Pb	V	Ni	Na	Cl
PM_{2.5}	1	0.26	0.11	0.48	0.21	0.29	0.29	0.48	0.80	0.15	0.12	0.16	0.33	0.18	0.23	0.08	0.48	-0.02
Al	0.26	1	0.97	0.62	0.60	0.88	0.61	0.65	0.35	0.04	0.01	-0.07	0.08	0.20	0.00	-0.09	0.38	0.04
Si	0.11	0.97	1	0.58	0.63	0.86	0.61	0.56	0.15	-0.03	0.02	-0.09	0.02	0.16	-0.01	-0.08	0.26	0.05
K	0.48	0.62	0.58	1	0.44	0.63	0.57	0.49	0.34	0.12	0.16	0.27	0.47	0.35	0.30	0.30	0.44	0.12
Ca	0.21	0.60	0.63	0.44	1	0.62	0.56	0.36	0.15	-0.05	0.16	0.06	0.06	0.14	0.22	0.03	0.29	0.21
Ti	0.29	0.88	0.86	0.63	0.62	1	0.69	0.58	0.31	0.02	0.10	0.03	0.18	0.17	0.09	-0.01	0.30	0.03
Fe	0.29	0.61	0.61	0.57	0.56	0.69	1	0.37	0.25	0.03	0.27	0.18	0.20	0.21	0.31	0.29	0.24	0.07
Mg	0.48	0.65	0.56	0.49	0.36	0.58	0.37	1	0.64	0.17	0.06	-0.01	0.26	0.33	0.10	-0.01	0.77	0.14
S	0.80	0.35	0.15	0.34	0.15	0.31	0.25	0.64	1	0.23	0.03	0.04	0.32	0.19	0.14	-0.02	0.65	-0.07
As	0.15	0.04	-0.03	0.12	-0.05	0.02	0.03	0.17	0.23	1	0.14	0.08	0.03	0.31	0.01	0.07	0.27	0.10
Cu	0.12	0.01	0.02	0.16	0.16	0.10	0.27	0.06	0.03	0.14	1	0.30	0.17	0.22	0.24	0.29	0.09	0.14
Zn	0.16	-0.07	-0.09	0.27	0.06	0.03	0.18	-0.01	0.04	0.08	0.30	1	0.21	0.32	0.36	0.38	0.04	0.06
Br	0.33	0.08	0.02	0.47	0.06	0.18	0.20	0.26	0.32	0.03	0.17	0.21	1	0.11	0.27	0.29	0.38	0.11
Pb	0.18	0.20	0.16	0.35	0.14	0.17	0.21	0.33	0.19	0.31	0.22	0.32	0.11	1	0.14	0.13	0.33	0.17
V	0.23	0.00	-0.01	0.30	0.22	0.09	0.31	0.10	0.14	0.01	0.24	0.36	0.27	0.14	1	0.84	0.31	0.12
Ni	0.08	-0.09	-0.08	0.30	0.03	-0.01	0.29	-0.01	-0.02	0.07	0.29	0.38	0.29	0.13	0.84	1	0.16	0.07
Na	0.48	0.38	0.26	0.44	0.29	0.30	0.24	0.77	0.65	0.27	0.09	0.04	0.38	0.33	0.31	0.16	1	0.37
Cl	-0.02	0.04	0.05	0.12	0.21	0.03	0.07	0.14	-0.07	0.10	0.14	0.06	0.11	0.17	0.12	0.07	0.37	1

Figure S1. Changes in mmHg and 95% CI in newborn systolic blood pressure for an IQR increase in the two to seven-days average of each PM_{2.5} mass components.



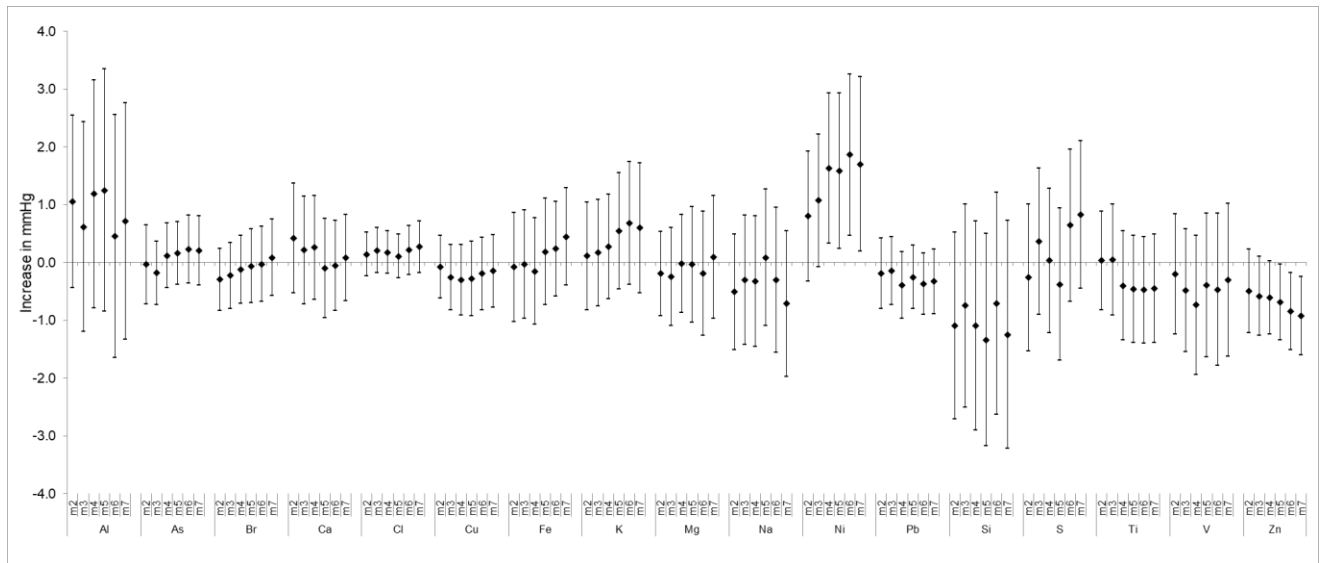
Results from the multi-pollutant model with all components in the same model. All models adjusted for maternal age, 3rd trimester BP, race/ethnicity, education, and income, infant age and state at BP, sine, cosine, temperature, birthweight and PM_{2.5}.

Figure S2. Changes in mmHg and 95% CI in newborn systolic blood pressure for an IQR increase in the two to seven-days average of each PM_{2.5} mass components.



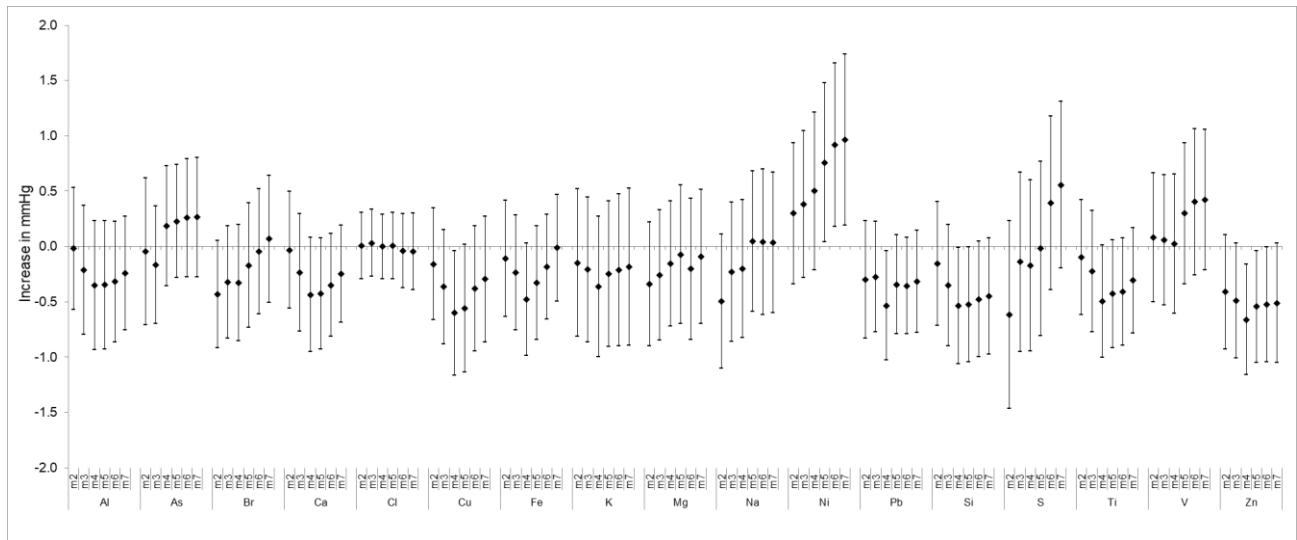
Results for the two-pollutant model including each individual PM_{2.5} component and PM_{2.5}. All models adjusted for maternal age, 3rd trimester BP, race/ethnicity, education, and income, infant age and state at BP, sine, cosine, temperature, birthweight and PM_{2.5}.

Figure S3. Changes in mmHg and 95% CI in newborn diastolic blood pressure for an IQR increase in the two to seven-days average of each PM_{2.5} mass components.



Results from the multi-pollutant model with all components in the same model. All models adjusted for maternal age, 3rd trimester BP, race/ethnicity, education, and income, infant age and state at BP, sine, cosine, temperature, birthweight and PM_{2.5}.

Figure S4. Changes in mmHg and 95% CI in newborn diastolic blood pressure for an IQR increase in the two to seven-days average of each PM_{2.5} mass components.



Results for the two-pollutant model including each individual PM_{2.5} component and PM_{2.5}. All models adjusted for maternal age, 3rd trimester BP, race/ethnicity, education, and income, infant age and state at BP, sine, cosine, temperature, birthweight and PM_{2.5}.